

Lab 1 - Knowledge Is Power

Paul Flynn

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Professor: Thomas Kennedy

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1 Introduction

1.1 Societal Problem

Young people ages 16–19 increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding their inner workings and the consequences that may ensue. As they gain more independence into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences. Despite this, law-related education remains largely absent from K-12 curricula, leaving young adults underprepared for real-world legal situations. Additionally, teen drivers face higher safety risks that are often linked to poor judgement and decision-making. [5.1, 5.2, 5.3, 5.4]

1.2 Needed Solution Characteristics

The platform should teach legal concepts through interactive activities that ask users to apply laws to realistic situations rather than simply memorize definitions. To remain accessible, it should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently. It should also explain the connection between each lesson and its official legal sources while introducing legal vocabulary in context so that users can recognize practical terms used across different legal topics.

1.3 Product Introduction

Knowledge Is Power (KIP) is a digital legal literacy platform designed to teach essential, everyday laws to young adults. It serves as a high-quality supplemental educational aid for parents, teachers, and school administrators. KIP's goal is to help young adults recognize legally significant situations and understand the potential consequences of their choices, to prepare them to make more informed decisions. The prototype will demonstrate these goals through a web-based service that supports user accounts, learning modules, interactive scenarios, progress tracking, and recommendation features based on user performance.

2 KIP (Knowledge Is Power) Product Description

KIP is structured as a web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. The system presents users with short misconception-based quizzes and longer branching scenarios, then uses their performance to recommend future practice. Its software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. The goal of these features is to assess users' current legal understanding, strengthen their recognition of legal situations, and reinforce informed decision-making through repeated low-stakes practice.

2.1 Key Product Features and Capabilities

KIP will have a variety of different intuitive ways for the user to interact with such as hands on scenarios, unique game interfaces, and progress tracking.

2.1.1 Solution Overview

KIP uses focused games, real-world examples, curated media, and official resources to educate young adults on practical legal boundaries.

2.1.2 Goals and Objectives

KIP aims to assess, improve, and reinforce legal knowledge among tech-friendly youth through repeatable, low-stakes interactive activities. Its user experience (UX) is designed to increase recognition of legal situations, improve decision-making skills, and strengthen long-term retention of legal concepts.

2.1.3 Interactive Gamification:

KIP uses focused, low-stakes games to improve users' confidence in making legal decisions.

2.1.3.1 "Fact or Fiction?" Game

This short-form game presents users with common legal beliefs and misconceptions, offering “true”, “false”, or “context-dependent” as answers. This approach will help users challenge assumptions, spot legal nuance, and build foundational knowledge to prepare them for longer and more elaborate scenarios.

2.1.3.2 "Choose Your Outcome" Game

This multiple-choice, long-form game encourages users to interact with people, objects, and documents to learn scenario details presented to them before making a decision. Every choice will demonstrate its consequences to illuminate its legal significance, which encourages users to slow down, evaluate context, and understand what information an outcome is dependent on.

2.1.4 Progress Tracking and Analytics

KIP tracks topical strengths, weaknesses, and completion during use. This information allows KIP to recommend relevant games and modules while helping each game session offer questions, scenarios, and legal concepts that meaningfully test the user.

2.2 Major Components (Hardware/Software)

KIP is designed with three main stacks for its functionality: Presentation layer, Application layer, and data layer. Each layer deals with different and unique features of KIP that affect the user’s experience. See Figure 1: Diagram of Major Software Components for a visual depiction of some of the interaction of the stack for KIP.

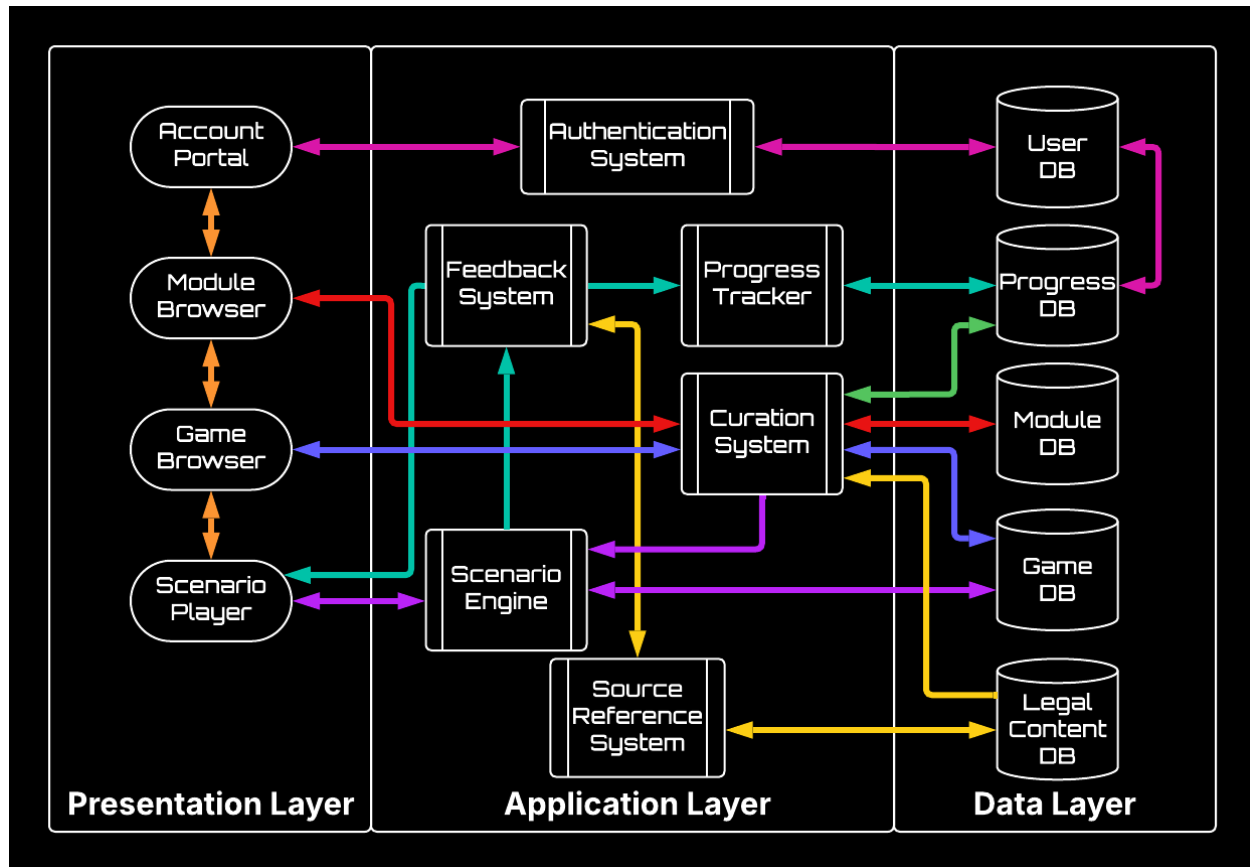


Figure 1: Diagram of Major Software Components

2.2.1 Presentation Layer

The Presentation Layer is the layer that the user will have direct interactions with. It contains the webpages and web applications that he or she will use with KIP.

2.2.1.1 Tech Stack

The technical code that is used for the Presentations Layer consists of the following languages and frameworks: JavaScript, Bootstrap, HTML5, and Angular.

2.2.1.2 Account Portal

This system allows established users to log in, and newcomers to create an account or play games without an account to get a taste of KIP, albeit with limited features.

2.2.1.3 Module Browser

This browser displays available legal learning modules such as traffic law, online conduct, school and work, and personal safety. It allows users to select a topic and view their progress status for that topic's module.

2.2.1.4 Game Browser

This browser displays the available games and prioritizes recommendations based on the user's progress, performance history, and demonstrated strengths or weaknesses in the legal subtopics covered by each game.

2.2.1.5 Scenario Player

This system is the sole interface users interact with while playing KIP's games. It presents interactive questions, branching scenarios, feedback, and source references during gameplay.

2.2.2 Application Layer

The Application Layer consists of business logic which handles any and all necessary application calls and methods.

2.2.2.1 Tech Stack

The technical code that is used for the Application Layer consists of the following languages and frameworks: Java, JDBC, and Spring Boot.

2.2.2.2 Authentication System

This system manages user login, account creation, guest access, and session validation. It determines whether a user is using KIP as a registered account or as a limited guest, and it controls which features are available based on that status.

2.2.2.3 Scenario Engine

This system loads and manages the interactive content used during KIP's games. For a play session, it assembles a "match" of play, with multiple "rounds." Each round establishes unique

choices, scenario branches, consequences, and game states that appear in sequence. This is primarily informed by two fed selected game and the user's previous actions.

2.2.2.4 Feedback System

This system provides explanations after user decisions. It identifies whether an answer is correct, incorrect, or context-dependent, explaining the legal reasoning behind the result in order to help the user understand why a specific choice was safe, risky, or incomplete.

2.2.2.5 Curation System

This system organizes and recommends learning content based on the user's progress and performance history. It determines what games, modules, and scenarios to prioritize, depending on where the user needs more practice, while also allowing users to freely browse and ultimately select any available content.

2.2.2.6 Source Reference System

This system connects game content and feedback to official legal sources, educational resources, and relevant vocabulary. It allows users to see where KIP's information comes from and, when desired, review the original legal language and documents behind a lesson. It also supports content maintenance by tracking changes to legislation, updating references when needed, and flagging KIP content that may be outdated so administrators can review it, revise it, or disable access to it.

2.2.3 Data Layer

The Data Layer handles all the databases and their specific schemas and operations.

2.2.3.1 Tech Stack

The technical code that is used for the Application Layer consists of the following languages and frameworks: MySQL and Docker.

2.2.3.2 Databases

User Database, Progress Database, Module Database, Game Database, Legal Content Database

2.2.3.3 Functions

Stores user account information; stores user progress records; stores learning modules and associated content; stores game scenario questions and gameplay information; stores legal and educational resources.

2.2.4 Hardware Layer

The Hardware Layer is the layer that an operator or administrator would interact with. It also describes the user's physical interaction via a client web browser.

2.2.4.1 Server-Side Components

A virtual private server hosts the Java application, manages server-side processing, and provides access to the SQLite database file and application resources.

2.2.4.2 Client-Side Components

Consumer devices with a modern web browser and internet connection

2.2.4.3 Functions

Runs Java application code, stores persistent data, processes user requests, and delivers the web-based interface to users.

3 Identification of Case Study

KIP is not a solution that is meant for all people to be able to use. It specifically targets young to middle-aged people that have experience with technology. To see the planned progress of the real world product to the planned prototype view Table 1: Real-World Product VS Prototype Table.

3.1 Primary Target Audience

KIP's primary target audience is young people aged 16–19 who are preparing to enter adulthood and navigate high-risk legal areas, such as driving, work, and substance use. This group is well

suited for KIP because its members are old enough for legal consequences to become personally significant, but still young enough to be especially receptive to education, habit formation, and the correction of misconceptions. By focusing on users at this transitional stage, KIP can help build foundational legal awareness and encourage informed decision-making before risky habits or misunderstandings become more difficult to change.

3.2 Secondary Stakeholders

Parents, teachers, and high school administrators may use KIP as a supplemental educational tool. These stakeholders benefit from KIP as a way to reinforce legal awareness outside the classroom.

3.3 Future Users

Future users of KIP may be members of juvenile diversion programs, community youth groups, and driving schools. These groups may use KIP as educational resources to improve legal literacy in structured settings.

Features & Functionality	RWP	Prototype (Planned)	Prototype (Actual)
United States Traffic Law	Coverage of Federal and all fifty states	Virginia traffic only	TBD
Copyright Law	Comprehensive copyright law	Social media platforms	TBD
Adaptive Game Difficulty	Relaxed, Standard, Intense modes	Level 1 (Relaxed)	TBD

		Level 2 (Standard) Level 3 (Intense)	
User Account and History	Server-side database	SQLite	TBD

Table 1: Real-World Product VS Prototype Table

4 Glossary

The following entries are some of the unique acronyms that are used in this document.

4.1 CDC

Centers for Disease Control and Prevention.

4.2 KIP

Knowledge Is Power.

4.3 MFCD

Major Functional Component Diagram.

4.4 UX

User Experience.

5 References

These references are some of the key points of information used while developing KIP.

5.1 Centers for Disease Control and Prevention. (2026)

<https://www.cdc.gov/transportation-safety/index.html>

5.2 Legal Services Corporation. (2017).

The justice gap: Measuring the unmet civil legal needs of low-income Americans.

<https://www.lsc.gov/sites/default/files/images/TheJusticeGap-FullReport.pdf>

5.3 National Highway Traffic Safety Administration. (2024).

Teen driving: Talk to your teen driver about safe driving. [https://www.nhtsa.gov/teen-](https://www.nhtsa.gov/teen-driving/parents-talk-your-teen-driver-about-safe-driving)

[driving/parents-talk-your-teen-driver-about-safe-driving](https://www.nhtsa.gov/teen-driving/parents-talk-your-teen-driver-about-safe-driving)

5.4 Touro Law Jacob D. Fuchsberg Law Center. (2017).

A Lawyer's Experience in K-12 Law-Related Education: Lessons and Opportunities.

<https://digitalcommons.tourolaw.edu/cgi/viewcontent.cgi?article=1023&context=jel>